

KSP Datathon 2026 — Challenge 1

Intelligent Conversational AI & Crime Analytics Platform

Data Architecture & Schema Reference

Document version	v1.0
Scope	Synthetic data generation — Phase 1 prototype
Tables defined	18 tables across 6 groups
Geospatial	Columns stored; spatial queries deferred to Phase 2
Legal framework	BNS 2023 / BNSS 2023 (post-July 2024) + IPC/CrPC legacy
Datathon modules covered	1, 2, 3, 4, 5, 6, 9, 10 (Modules 7 & 8 deferred)

1. Purpose & Scope

This document defines all data sources, table schemas, primary keys, foreign keys, and inter-table relationships required to build the synthetic dataset for the KSP Challenge 1 prototype. It is intended to be carried into the data-engineering conversation as a self-contained reference.

Key assumptions made in this design:

- All data is synthetically generated — no real KSP dataset is provided.
- All structured fields are assumed available; free-text narrative fields (FIR complaint text, case diary entries, MO descriptions) are included and will be indexed for semantic search (C7 / RAG layer).
- Coordinates (lat/lng) are stored as plain DECIMAL columns on all relevant tables. PostGIS spatial extensions and map rendering are deferred to Phase 2.
- The BNS (Bharatiya Nyaya Sanhita) came into force on July 1, 2024. FIRs registered before this date use IPC sections; after this date use BNS sections. Both are stored on the FIR record.
- No full Aadhaar numbers, biometric data, or sensitive PII beyond what appears in a real FIR is stored.

2. Data Sources Required

Eighteen tables across six logical groups cover the full v1 user-story backlog (23 stories). The table below maps each group to the stories it enables.

Group A — Jurisdiction Masters	Station hierarchy, officer records — foundation for all role-scoped queries (Stories 14, 15, 20)
Group B — Lookup / Reference	Crime types, BNS/IPC sections, MO codes — powers the semantic layer C5 (Stories 1, 2, 5, 7, 19)
Group C — Persons	Master entity for accused, victims, witnesses, complainants — foundation for entity resolution C8 (Stories 1, 4, 12, 17, 18)
Group D — FIR & Investigation	Core transactional tables — FIRs, case diaries, arrests, chargesheets — powers C6, C7, C12 (Stories 1–8, 10, 16, 18, 19, 21, 23)
Group E — Criminal Intelligence	History sheets, gangs, known associates — powers C9 graph engine (Stories 4, 8, 12, 17)
Group F — Property & Vehicles	Stolen property, vehicle registry — powers field lookup (Story 16)
Group G — External / Socio-Economic	District census indicators, events calendar — powers C17 (Stories 13, 21, 24)
Group H — Audit	Query audit log — powers C16 (Stories 15, 26)

3. Table Schemas

Group A — Jurisdiction & Officer Masters

These tables form the geographic and organizational spine of the system. Every FIR, officer, and query is scoped against this hierarchy.

A1. districts

Field Name	Type	Constraint	Notes / Description
<code>district_id</code>	CHAR(3)	PRIMARY KEY	e.g. 'BLR', 'MYS', 'DWD', 'HBL', 'BJP'
<code>district_name</code>	VARCHAR(100)	NOT NULL	English name — e.g. 'Bengaluru Urban'
<code>district_name_kn</code>	VARCHAR(100)		Kannada script — e.g. 'ಬೆಂಗಳೂರು ನಗರ'
<code>division</code>	VARCHAR(50)		Administrative division — South / North / Central / Eastern / Western
<code>hq_city</code>	VARCHAR(100)		Headquarters city name

A2. sub_divisions

Field Name	Type	Constraint	Notes / Description
<code>subdivision_id</code>	CHAR(6)	PRIMARY KEY	e.g. 'BLR-N1'
<code>subdivision_name</code>	VARCHAR(100)	NOT NULL	
<code>district_id</code>	CHAR(3)	FK → districts	

A3. circles

Field Name	Type	Constraint	Notes / Description
<code>circle_id</code>	CHAR(8)	PRIMARY KEY	
<code>circle_name</code>	VARCHAR(100)	NOT NULL	
<code>subdivision_id</code>	CHAR(6)	FK → sub_divisions	

A4. police_stations

Field Name	Type	Constraint	Notes / Description
<code>station_id</code>	CHAR(10)	PRIMARY KEY	Format: KA-[DIST]-[3-digit serial] e.g. KA-MYS-012
<code>station_name</code>	VARCHAR(150)	NOT NULL	e.g. 'Jayanagar Police Station'
<code>station_name_kn</code>	VARCHAR(150)		Kannada script name
<code>circle_id</code>	CHAR(8)	FK → circles	
<code>district_id</code>	CHAR(3)	FK → districts	Denormalised for fast district-level filtering
<code>station_type</code>	VARCHAR(30)		City / Rural / Town / Traffic / Women / Cyber
<code>address</code>	TEXT		
<code>latitude</code>	DECIMAL(9,6)		Stored now; spatial queries deferred to Phase 2
<code>longitude</code>	DECIMAL(9,6)		Stored now; spatial queries deferred to Phase 2
<code>phone</code>	VARCHAR(15)		
<code>established_year</code>	SMALLINT		

A5. officers

Field Name	Type	Constraint	Notes / Description
<code>officer_id</code>	VARCHAR(15)	PRIMARY KEY	Belt/service number e.g. 'KSP-12345'
<code>name</code>	VARCHAR(150)	NOT NULL	
<code>name_kn</code>	VARCHAR(150)		Kannada script
<code>rank</code>	VARCHAR(50)	NOT NULL	Constable / HC / ASI / SI / PI / Inspector / DSP / SP / DCP / SSP / DIG / IG / ADGP / DGP
<code>rank_code</code>	VARCHAR(10)		Short code e.g. 'SI', 'PI', 'DSP'
<code>station_id</code>	CHAR(10)	FK → police_stations	Current posting
<code>district_id</code>	CHAR(3)	FK → districts	Denormalised for fast scoping
<code>role</code>	VARCHAR(30)	NOT NULL	IO / SHO / Circle_Inspector / Supervisor / Analyst / Admin

Field Name	Type	Constraint	Notes / Description
date_of_joining	DATE		
is_active	BOOLEAN	DEFAULT TRUE	
phone	VARCHAR(15)		
email	VARCHAR(100)		

Group B — Lookup / Reference Tables

These tables form the domain semantic layer (C5). They must be populated with real BNS/IPC data before synthetic FIRs are generated, since FIRs reference section IDs.

B1. bns_sections

Field Name	Type	Constraint	Notes / Description
section_id	VARCHAR(10)	PRIMARY KEY	e.g. 'BNS-303', 'IPC-379'
bns_section	VARCHAR(20)	NOT NULL	Section number e.g. '303'
bns_description	TEXT	NOT NULL	Full offence description
ipc_equivalent	VARCHAR(20)		IPC section number for pre-July-2024 FIRs — NULL if no equivalent
ipc_description	TEXT		
offence_category	VARCHAR(50)		Property / Person / Sexual / Economic / Public_Order / Narcotics / Cyber
is_cognizable	BOOLEAN	NOT NULL	Determines whether FIR is mandatory
is_bailable	BOOLEAN	NOT NULL	
max_punishment_yrs	SMALLINT		For BNSS chargesheet deadline calculation
chargesheet_days	SMALLINT	NOT NULL	60 for most offences; 90 for death/life imprisonment offences per BNSS

B2. crime_types

Field Name	Type	Constraint	Notes / Description
crime_type_id	VARCHAR(20)	PRIMARY KEY	e.g. 'THEFT-VEHICLE', 'ROBBERY-CHAIN', 'ASSAULT-GRIEVOUS'
crime_type_name	VARCHAR(100)	NOT NULL	English label
crime_type_name_kn	VARCHAR(100)		Kannada label — ಕಳ್ಳತನ, ದರೋಡೆ, ಹಲ್ಲೆ
parent_category	VARCHAR(50)	NOT NULL	Theft / Robbery / Assault / Homicide / Sexual / Economic / Cyber / Narcotics / Public_Order
primary_bns_section	VARCHAR(10)	FK → bns_sections	Most common section applied for this crime type
severity_level	SMALLINT	NOT NULL	1 (petty) to 5 (heinous) — used in rule-based risk scoring

B3. mo_codes

i The MO (Modus Operandi) library is the machine-readable version of what the BPR&D Modus Operandi Bureau maintains. Seed with ~80–100 realistic entries.

Field Name	Type	Constraint	Notes / Description
mo_code_id	VARCHAR(15)	PRIMARY KEY	e.g. 'MO-THEFT-001', 'MO-ROB-005'
crime_type_id	VARCHAR(20)	FK → crime_types	
mo_description	TEXT	NOT NULL	e.g. 'Accused enters via unlocked rear door during night hours; removes jewellery from bedroom'
mo_description_kn	TEXT		Kannada translation
target_type	VARCHAR(50)		Residential / Commercial / Vehicle / Person / ATM / Temple
time_pattern	VARCHAR(50)		Night / Early_Morning / Business_Hours / Evening / Variable
tool_used	VARCHAR(100)		e.g. 'Iron rod', 'Duplicate key', 'Knife', 'None / Physical force'
typical_gang_size	SMALLINT		1 for solo offenders; 2–8 for gangs

Group C — Persons (Master Entity)

The persons table is the most important table in the system. Every accused, victim, witness, and complainant is a person record. Entity resolution (C8) operates on this table to deduplicate individuals who appear under different name variants, scripts, or spellings across FIRs and stations.

C1. persons

Field Name	Type	Constraint	Notes / Description
person_id	UUID	PRIMARY KEY	Internal canonical ID — generated at ingestion
full_name	VARCHAR(200)	NOT NULL	English transliteration
full_name_kn	VARCHAR(200)		Kannada script name
aliases	TEXT[]		Array of all known name variants e.g. ['Ravi', 'Ravi Kumar N', 'ರವಿ ಕುಮಾರ್']
gender	VARCHAR(10)	NOT NULL	Male / Female / Other / Unknown
date_of_birth	DATE		NULL if unknown
age_approx	SMALLINT		Used when DOB is unknown — estimated at time of FIR
father_name	VARCHAR(200)		Critical for disambiguation — many common names share first names
mother_name	VARCHAR(200)		
spouse_name	VARCHAR(200)		
nationality	VARCHAR(50)	DEFAULT 'Indian'	
religion	VARCHAR(50)		For sociological Module 4 analysis — stored, not displayed by default
caste	VARCHAR(100)		For Module 4 — socio-economic correlation analysis
education_level	VARCHAR(50)		Illiterate / Primary / Secondary / Graduate / Postgraduate / Unknown
occupation	VARCHAR(100)		Free text — e.g. 'Daily wage labourer', 'Auto driver', 'Contractor'
occupation_category	VARCHAR(50)		Unemployed / Daily_Wage / Business / Service / Student / Agricultural / Other
monthly_income_approx	INT		Approximate monthly income in INR
address_current	TEXT		Last known current address
address_permanent	TEXT		Native place / permanent address
district_id	CHAR(3)	FK → districts	Home district — for jurisdictional analytics
latitude	DECIMAL(9,6)		Home address coordinates — nullable; Phase 2 use
longitude	DECIMAL(9,6)		
aadhaar_last4	CHAR(4)		Last 4 digits only — no full Aadhaar stored
phone_primary	VARCHAR(15)		Primary phone number
phone_secondary	VARCHAR(15)		
is_history_sheeted	BOOLEAN	DEFAULT FALSE	Synced from history_sheets table
is_proclaimed_offender	BOOLEAN	DEFAULT FALSE	Court-declared absconder
photo_id	VARCHAR(100)		Reference to photo store (path/URL) — not stored inline
fingerprint_id	VARCHAR(100)		Reference to biometric store — not stored inline
er_cluster_id	UUID		Entity resolution — links records believed to be the same person
er_confidence	DECIMAL(4,3)		Confidence score 0.000–1.000 that this is the canonical record
er_status	VARCHAR(20)		Canonical / Duplicate / Pending_Review
created_at	TIMESTAMP	DEFAULT NOW()	
updated_at	TIMESTAMP		

C2. person_phones

❖ Separate table for phone number history. Critical for network analysis — shared phone numbers link accused across FIRs.

Field Name	Type	Constraint	Notes / Description
phone_record_id	UUID	PRIMARY KEY	
person_id	UUID	FK → persons	
phone_number	VARCHAR(15)	NOT NULL	Normalised to +91XXXXXXXXXX format

Field Name	Type	Constraint	Notes / Description
phone_type	VARCHAR(20)		Mobile / Landline / WhatsApp
is_active	BOOLEAN		Whether number is currently in use
first_seen_date	DATE		Date first appeared in any FIR
last_seen_date	DATE		
source_fir_id	VARCHAR(20)	FK → firs	FIR where this number first appeared

C3. person_addresses

Field Name	Type	Constraint	Notes / Description
address_id	UUID	PRIMARY KEY	
person_id	UUID	FK → persons	
address_text	TEXT	NOT NULL	Full address text
address_type	VARCHAR(20)	NOT NULL	Current / Permanent / Work / Associate
station_id	CHAR(10)	FK → police_stations	Jurisdictional station for this address
latitude	DECIMAL(9,6)		Phase 2
longitude	DECIMAL(9,6)		Phase 2
valid_from	DATE		
valid_to	DATE		NULL = currently valid

Group D — FIR & Investigation Tables

The core transactional spine of the system. Every investigative story (1–6, 10, 16, 18, 19, 21) draws from this group. The `firs` table is the anchor — all other tables in this group are children of it.

D1. `firs`

Field Name	Type	Constraint	Notes / Description
<code>fir_id</code>	VARCHAR(20)	PRIMARY KEY	Format: KA-[DIST]-[STN]-[YEAR]-[SERIAL] e.g. KA-MYS-012-2024-0347
<code>station_id</code>	CHAR(10)	FK → <code>police_stations</code>	Station where FIR is registered
<code>district_id</code>	CHAR(3)	FK → <code>districts</code>	Denormalised for fast district filtering
<code>fir_number</code>	VARCHAR(10)	NOT NULL	Local serial number within station-year e.g. '0347'
<code>fir_year</code>	SMALLINT	NOT NULL	
<code>registration_date</code>	TIMESTAMP	NOT NULL	Date and time FIR was registered
<code>incident_date</code>	TIMESTAMP		When the crime occurred — may differ from registration
<code>incident_date_approx</code>	BOOLEAN	DEFAULT FALSE	TRUE if victim/witness could not give exact time
<code>incident_location</code>	TEXT	NOT NULL	Free text location description — indexed for RAG
<code>incident_area</code>	VARCHAR(100)		Locality/area name e.g. 'Jayanagar 4th Block'
<code>incident_latitude</code>	DECIMAL(9,6)		Phase 2 spatial queries
<code>incident_longitude</code>	DECIMAL(9,6)		Phase 2 spatial queries
<code>crime_type_id</code>	VARCHAR(20)	FK → <code>crime_types</code>	
<code>primary_bns_section</code>	VARCHAR(10)	FK → <code>bns_sections</code>	Principal section applied
<code>additional_sections</code>	TEXT[]		Array of additional BNS section IDs
<code>ipc_sections</code>	TEXT[]		Array of IPC section numbers — populated for pre-July-2024 FIRs
<code>is_pre_bns</code>	BOOLEAN	DEFAULT FALSE	TRUE if registered before July 1, 2024 — governs which sections apply
<code>complaint_text</code>	TEXT		Complainant's statement verbatim — PRIMARY RAG index field
<code>complaint_text_kn</code>	TEXT		Kannada version
<code>fir_narrative</code>	TEXT		IO's written FIR summary — RAG indexed
<code>mo_code_id</code>	VARCHAR(15)	FK → <code>mo_codes</code>	Matched MO code from library
<code>mo_description_free</code>	TEXT		Free-text MO detail when no code matches — RAG indexed
<code>weapons_used</code>	TEXT[]		Array of weapons/tools mentioned
<code>property_involved</code>	BOOLEAN	DEFAULT FALSE	Whether stolen property records exist
<code>complainant_id</code>	UUID	FK → <code>persons</code>	The person who filed the complaint
<code>io_officer_id</code>	VARCHAR(15)	FK → <code>officers</code>	Investigating Officer assigned
<code>sho_officer_id</code>	VARCHAR(15)	FK → <code>officers</code>	Station House Officer at time of registration
<code>investigation_status</code>	VARCHAR(30)	NOT NULL	Registered / Under_Investigation / Chargesheet_Filed / Closed_True / Closed_False / Referred
<code>chargesheet_deadline</code>	DATE		Computed: <code>registration_date</code> + <code>chargesheet_days</code> from <code>bns_sections</code>
<code>fir_type</code>	VARCHAR(20)		Suo_Motu / Complaint / Transfer / Court_Direction / Zero_FIR
<code>is_zero_fir</code>	BOOLEAN	DEFAULT FALSE	TRUE if registered at non-jurisdictional station and transferred
<code>created_at</code>	TIMESTAMP	DEFAULT NOW()	
<code>updated_at</code>	TIMESTAMP		

D2. `fir_accused`

Field Name	Type	Constraint	Notes / Description
<code>fir_accused_id</code>	UUID	PRIMARY KEY	
<code>fir_id</code>	VARCHAR(20)	FK → <code>firs</code>	
<code>person_id</code>	UUID	FK → <code>persons</code>	
<code>role</code>	VARCHAR(30)	NOT NULL	Main_Accused / Co_Accused / Absconder / Unknown / Suspect
<code>accused_serial</code>	SMALLINT	NOT NULL	A1, A2, A3 within the FIR
<code>is_arrested</code>	BOOLEAN	DEFAULT FALSE	
<code>arrest_date</code>	DATE		

Field Name	Type	Constraint	Notes / Description
is_juvenile	BOOLEAN	DEFAULT FALSE	Governs JJB referral

D3. fir_victims

Field Name	Type	Constraint	Notes / Description
fir_victim_id	UUID	PRIMARY KEY	
fir_id	VARCHAR(20)	FK → firs	
person_id	UUID	FK → persons	
victim_serial	SMALLINT	NOT NULL	V1, V2 within the FIR
injury_type	VARCHAR(50)		None / Minor / Grievous / Fatal
victim_statement	TEXT		RAG indexed
victim_statement_kn	TEXT		Kannada version

D4. fir_witnesses

Field Name	Type	Constraint	Notes / Description
fir_witness_id	UUID	PRIMARY KEY	
fir_id	VARCHAR(20)	FK → firs	
person_id	UUID	FK → persons	
witness_type	VARCHAR(30)		Eyewitness / Panch / Expert / Character / Mahazar
statement_recorded	BOOLEAN	DEFAULT FALSE	
statement_date	DATE		

D5. case_diary_entries

i The case diary is a legally mandated daily record of investigation. Free-text entries are the richest source of investigative narrative and are a primary RAG index target.

Field Name	Type	Constraint	Notes / Description
entry_id	UUID	PRIMARY KEY	
fir_id	VARCHAR(20)	FK → firs	
officer_id	VARCHAR(15)	FK → officers	Officer making the entry
entry_date	DATE	NOT NULL	
entry_number	SMALLINT	NOT NULL	Sequential within case — UNIQUE (fir_id, entry_number)
entry_text	TEXT	NOT NULL	PRIMARY RAG index field — full diary entry text
entry_text_kn	TEXT		Kannada version
action_taken	VARCHAR(100)		Spot_Visit / Arrest / Statement / Seizure / FSL_Sent / Court_Production / Other
next_action_planned	TEXT		IO's stated next step
created_at	TIMESTAMP	DEFAULT NOW()	

D6. arrests

Field Name	Type	Constraint	Notes / Description
arrest_id	UUID	PRIMARY KEY	
fir_accused_id	UUID	FK → fir_accused	
fir_id	VARCHAR(20)	FK → firs	Denormalised for fast lookup
person_id	UUID	FK → persons	Denormalised for fast lookup
arresting_officer_id	VARCHAR(15)	FK → officers	
arrest_date	TIMESTAMP	NOT NULL	
arrest_location	TEXT		
arrest_latitude	DECIMAL(9,6)		Phase 2
arrest_longitude	DECIMAL(9,6)		Phase 2
remand_type	VARCHAR(20)		Police_Custody / Judicial_Custody
remand_expiry	DATE		
bail_status	VARCHAR(20)		Not_Applied / Applied / Granted / Rejected
bail_date	DATE		
is_absconding	BOOLEAN	DEFAULT FALSE	

D7. chargesheets

Field Name	Type	Constraint	Notes / Description
chargesheet_id	UUID	PRIMARY KEY	
fir_id	VARCHAR(20)	FK → firs UNIQUE	One chargesheet per FIR (supplementary via self-ref)
filing_officer_id	VARCHAR(15)	FK → officers	
filing_date	DATE	NOT NULL	
court_name	VARCHAR(150)		e.g. 'JMFC Court, Mysuru'
court_case_number	VARCHAR(50)		CC/SC number assigned by court
sections_applied	TEXT[]	NOT NULL	Final BNS sections as filed
sections_applied_ipc	TEXT[]		IPC sections for pre-July-2024 history in the case
num_accused	SMALLINT		
num_witnesses	SMALLINT		
summary_text	TEXT		IO's case summary — RAG indexed; source for Story 19 brief
filing_status	VARCHAR(20)		Filed / Returned / Supplementary
is_supplementary	BOOLEAN	DEFAULT FALSE	
parent_chargesheet_id	UUID	FK → chargesheets	Self-referential for supplementary chargesheets

D8. court_disposals

i Lightweight outcome table — enough to power 'similar past cases and their outcomes' (Story 2) without modelling the full court process.

Field Name	Type	Constraint	Notes / Description
disposal_id	UUID	PRIMARY KEY	
chargesheet_id	UUID	FK → chargesheets	
fir_id	VARCHAR(20)	FK → firs	Denormalised
disposal_date	DATE		
court_name	VARCHAR(150)		
outcome	VARCHAR(30)	NOT NULL	Convicted / Acquitted / Compounded / Withdrawn / Pending
sentence_details	TEXT		If convicted — e.g. '3 years RI + ₹5000 fine'
notes	TEXT		

Group E — Criminal Intelligence Tables

These tables power the graph engine (C9) and the repeat-offender stories (4, 8, 12, 17). The known_associates table is the edge list for the criminal network graph.

E1. history_sheets

i History sheets (rowdy sheets) are Karnataka's habitual offender register. Only persons with is_history_sheeted = TRUE in persons have a record here.

Field Name	Type	Constraint	Notes / Description
history_sheet_id	UUID	PRIMARY KEY	
person_id	UUID	FK → persons UNIQUE	One history sheet per person
hs_number	VARCHAR(20)	NOT NULL	Station-assigned HS number e.g. 'MYS-HS-0234'
station_id	CHAR(10)	FK → police_stations	Originating station
opened_date	DATE	NOT NULL	
opened_by_officer	VARCHAR(15)	FK → officers	
category	VARCHAR(30)	NOT NULL	Rowdy / Goonda / History_Sheeteer / Bootlegger / Gambler / Other
is_active	BOOLEAN	DEFAULT TRUE	
last_reviewed_date	DATE		BNSS requires periodic review
review_notes	TEXT		
risk_level	VARCHAR(10)	NOT NULL	Low / Medium / High / Very_High — rule-based; powers Story 12
total_cases_count	SMALLINT	DEFAULT 0	Denormalised count — updated on each new FIR linkage
conviction_count	SMALLINT	DEFAULT 0	
absconding_count	SMALLINT	DEFAULT 0	
last_crime_date	DATE		Date of most recent FIR as accused

E2. history_sheet_entries

Field Name	Type	Constraint	Notes / Description
hs_entry_id	UUID	PRIMARY KEY	
history_sheet_id	UUID	FK → history_sheets	
entry_date	DATE	NOT NULL	
entry_text	TEXT	NOT NULL	Officer's narrative entry — RAG indexed
officer_id	VARCHAR(15)	FK → officers	
related_fir_id	VARCHAR(20)	FK → firs	Nullable — not all entries link to FIRs

E3. gangs

Field Name	Type	Constraint	Notes / Description
gang_id	UUID	PRIMARY KEY	
gang_name	VARCHAR(150)		e.g. 'Mysuru Chain Gang', 'KR Pete Auto Lifters'
gang_name_kn	VARCHAR(150)		Kannada script
primary_crime_type	VARCHAR(20)	FK → crime_types	
operating_district	CHAR(3)	FK → districts	Primary operating area
is_active	BOOLEAN	DEFAULT TRUE	
formation_approx_year	SMALLINT		
known_strength	SMALLINT		Approximate member count
notes	TEXT		Intelligence notes — RAG indexed

E4. gang_memberships

Field Name	Type	Constraint	Notes / Description
membership_id	UUID	PRIMARY KEY	
gang_id	UUID	FK → gangs	
person_id	UUID	FK → persons	

Field Name	Type	Constraint	Notes / Description
role_in_gang	VARCHAR(30)	NOT NULL	Leader / Sub-Leader / Member / Associate / Informant / Financier
joined_approx_date	DATE		
is_active	BOOLEAN	DEFAULT TRUE	
left_date	DATE		
notes	TEXT		

E5. known_associates

i Self-referential many-to-many on persons. This is the edge list for the criminal network graph (C9). A constraint (*person_id_a* < *person_id_b*) prevents duplicate pairs.

Field Name	Type	Constraint	Notes / Description
association_id	UUID	PRIMARY KEY	
person_id_a	UUID	FK → persons	Always the lower UUID — enforced by CHECK constraint
person_id_b	UUID	FK → persons	Always the higher UUID
association_type	VARCHAR(30)	NOT NULL	Co_Accused / Family / Employer_Employee / Neighbour / Gang_Member / Phone_Contact / Financier / Known_Receiver
first_seen_fir_id	VARCHAR(20)	FK → firs	FIR where this association was first recorded
confidence	VARCHAR(10)	NOT NULL	Confirmed / Suspected
notes	TEXT		

Group F — Property & Vehicles

F1. stolen_property

Field Name	Type	Constraint	Notes / Description
property_id	UUID	PRIMARY KEY	
fir_id	VARCHAR(20)	FK → firs	FIR in which this property was reported stolen
property_type	VARCHAR(50)	NOT NULL	Mobile / Jewellery / Vehicle / Cash / Electronics / Documents / Other
description	TEXT	NOT NULL	e.g. 'Gold chain 20g, Thali pattern' — RAG indexed
description_kn	TEXT		Kannada description
estimated_value_inr	INT		Estimated value at time of theft
serial_number	VARCHAR(100)		IMEI, serial number, hallmark — key for matching recovered property
is_recovered	BOOLEAN	DEFAULT FALSE	
recovery_date	DATE		
recovery_fir_id	VARCHAR(20)	FK → firs	FIR that led to recovery — links theft to recovery case

F2. vehicles

Field Name	Type	Constraint	Notes / Description
vehicle_id	UUID	PRIMARY KEY	
registration_number	VARCHAR(15)	NOT NULL UNIQUE	e.g. KA-09-AB-1234 — primary lookup field for Story 16
vehicle_type	VARCHAR(30)	NOT NULL	Two_Wheeler / Car / Auto / Truck / Bus / Other
make	VARCHAR(50)		Honda / TVS / Maruti / Bajaj / Royal Enfield / etc.
model	VARCHAR(50)		Activa / Pulsar / Swift / etc.
color	VARCHAR(30)		
year_of_manufacture	SMALLINT		
owner_person_id	UUID	FK → persons	Registered owner — may differ from current user
is_stolen	BOOLEAN	DEFAULT FALSE	
stolen_fir_id	VARCHAR(20)	FK → firs	
theft_date	DATE		
is_recovered	BOOLEAN	DEFAULT FALSE	
recovery_date	DATE		

Group G — External & Socio-Economic Data

These tables enable the sociological analysis stories (13, 24) and bandobast planning (21). For v1, all external data is at district level. Ward-level granularity is Phase 2.

G1. district_socioeconomic

i Seed with district-level data from Census 2011, NCRB Crime in India reports, and Karnataka Budget documents. Label source explicitly for credibility.

Field Name	Type	Constraint	Notes / Description
record_id	UUID	PRIMARY KEY	
district_id	CHAR(3)	FK → districts	
data_year	SMALLINT	NOT NULL	UNIQUE with district_id
population	INT		
population_urban_pct	DECIMAL(5,2)		% of population in urban areas
literacy_rate	DECIMAL(5,2)		
unemployment_rate	DECIMAL(5,2)		
per_capita_income_inr	INT		Annual per capita income
poverty_rate	DECIMAL(5,2)		% below poverty line
migrant_population_pct	DECIMAL(5,2)		% of population who are migrants
police_stations_count	SMALLINT		
officers_per_lakh_pop	DECIMAL(6,2)		Policing density metric
data_source	VARCHAR(100)	NOT NULL	e.g. 'Census 2011', 'NCRB Crime in India 2023'

G2. events_calendar

i Seed with major Karnataka festivals, elections, and recurring events. Powers Story 21 (bandobast planning) — IO can query 'historical incidents at Dasara, Mysuru' and see past patterns.

Field Name	Type	Constraint	Notes / Description
event_id	UUID	PRIMARY KEY	
event_name	VARCHAR(150)	NOT NULL	e.g. 'Dasara Procession', 'Karaga Festival', 'Belagavi Assembly Session'
event_name_kn	VARCHAR(150)		Kannada name
event_type	VARCHAR(30)	NOT NULL	Festival / Election / Strike / Sports / Religious / Fair / VIP_Visit
district_id	CHAR(3)	FK → districts	NULL for state-wide events
station_id	CHAR(10)	FK → police_stations	NULL for district-wide events
event_date_start	DATE	NOT NULL	
event_date_end	DATE		NULL for single-day events
expected_footfall	INT		Approximate crowd size
historical_incident_count	SMALLINT		Number of crime incidents recorded in prior years during this event
notes	TEXT		Historical incident patterns — RAG indexed

Group H — Audit & Governance

H1. query_audit_log

i Every query made through the platform is logged here — immutably. This table powers Stories 15 (full audit trail) and 26 (proactive anomaly detection). It is append-only; no UPDATE or DELETE is permitted.

Field Name	Type	Constraint	Notes / Description
log_id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	
officer_id	VARCHAR(15)	FK → officers	Who made the query
session_id	UUID	NOT NULL	Groups queries within a single login session
query_timestamp	TIMESTAMP	NOT NULL DEFAULT NOW()	
query_text	TEXT	NOT NULL	Exact natural-language query as submitted
query_language	VARCHAR(10)		'en' or 'kn'
query_type	VARCHAR(30)		Lookup / Network / Pattern / Summary / Export / Watchlist
records_accessed	TEXT[]		Array of FIR IDs and person UUIDs accessed in generating the answer
response_summary	TEXT		First 500 characters of the answer returned
components_used	TEXT[]		Architecture components invoked e.g. ['C6','C8','C12']
is_flagged	BOOLEAN	DEFAULT FALSE	Set by rule-based anomaly detection (Story 26)
flag_reason	VARCHAR(100)		Bulk_Lookup / Off_Hours_No_Case_Link / Repeated_Private_Citizen / Cross_District_No_Auth
reviewed_by	VARCHAR(15)	FK → officers	Admin who reviewed the flag
reviewed_at	TIMESTAMP		

4. Entity-Relationship Overview

The hierarchy below describes the primary relationships. Foreign key directions follow the arrow.

Jurisdiction Hierarchy

```
districts
└─ sub_divisions (district_id → districts)
   └─ circles (subdivision_id → sub_divisions)
      └─ police_stations (circle_id → circles)
         └─ officers (station_id → police_stations)
            └─ firs (station_id → police_stations)
```

Persons — the Master Entity

```
persons
└─ person_phones (person_id → persons) [1:many]
└─ person_addresses (person_id → persons) [1:many]
└─ history_sheets (person_id → persons) [1:1 – habitual offenders only]
   └─ history_sheet_entries (history_sheet_id → history_sheets) [1:many]
└─ gang_memberships (person_id → persons) [many:many → gangs]
└─ known_associates (person_id_a, person_id_b → persons) [self-ref many:many]
```

FIR — the Transaction Anchor

```
firs
└─ fir_accused (fir_id → firs, person_id → persons) [many:many]
   └─ arrests (fir_accused_id → fir_accused) [1:many]
└─ fir_victims (fir_id → firs, person_id → persons) [many:many]
└─ fir_witnesses (fir_id → firs, person_id → persons) [many:many]
└─ case_diary_entries (fir_id → firs) [1:many]
└─ stolen_property (fir_id → firs) [1:many]
└─ chargesheets (fir_id → firs) [1:1]
   └─ court_disposals (chargesheet_id → chargesheets) [1:1]
```

Reference / Lookup

crime_types	→ bns_sections	(primary_bns_section → bns_sections)
mo_codes	→ crime_types	(crime_type_id → crime_types)
firs	→ crime_types	(crime_type_id → crime_types)
firs	→ bns_sections	(primary_bns_section → bns_sections)
firs	→ mo_codes	(mo_code_id → mo_codes)
gangs	→ crime_types	(primary_crime_type → crime_types)

External Data

district_socioeconomic	→ districts	(district_id → districts)
events_calendar	→ districts	(district_id → districts, nullable)
events_calendar	→ police_stations	(station_id → police_stations, nullable)
query_audit_log	→ officers	(officer_id → officers)

5. Database Technology Decisions

Primary Store: PostgreSQL

- Handles all 18 relational tables.
- TEXT[] arrays cleanly cover multi-section FIRs, aliases, and component logs.
- UUID primary keys throughout — essential for synthetic data generation and future merging.
- DECIMAL(9,6) lat/lng columns stored inert in v1. PostGIS added later as CREATE EXTENSION — no data migration required.
- tsvector columns on complaint_text, fir_narrative, entry_text, mo_description_free, summary_text, and notes fields enable in-database full-text search as a fallback before the vector store is ready.

Supplementary: pgvector (on same Postgres instance)

- Embeds all RAG-indexed text fields using a multilingual model — recommended: intfloat/multilingual-e5-base (handles English + Kannada).
- Stores embeddings in a document_chunks table alongside source record ID and field name.
- Keeps the prototype stack to one database technology — significant operational simplification.

Graph: SQL-first, Neo4j optional

- known_associates, gang_memberships, and fir_accused give you a full graph in SQL.
- For the v1 demo, materialize the network using Python NetworkX at query time; render with vis.js or Cytoscape.js.
- Migrate to Neo4j only if graph traversal queries become too slow — unlikely at prototype scale.

6. Synthetic Data Generation Guidance

Target Volume

Districts	10 (mix of urban: BLR, MYS, HBL; rural: Kodagu, Chikkamagaluru, Raichur)
Police stations	50 (3–7 per district; use real KSP station names)
Officers	100 (mix of ranks; assign IOs, SHOs, supervisors)
Persons	500 (of whom ~150 are accused with criminal history)
FIRs	800 (spanning 2021–2025; mix of pre-BNS and post-BNS)
Case diary entries	2,000 (avg 2.5 per FIR)
Chargesheets	200 (25% of FIRs progressed to chargesheet)
Court disposals	80 (40% of chargesheets disposed)
Gangs	5 (8–15 members each; spread across districts)
Known associates	300 pairs
Stolen property	150 records
Vehicles	80 (of which 30 are flagged stolen)

Credibility Rules

- Names: use real Karnataka name patterns — Ramesh, Savitha, Nagaraj, Kavitha, Venkatesh, Manjunath, Shivakumar, Lakshmi. Surnames: Gowda, Reddy, Rao, Naik, Kumar, Swamy, Hegde, Shetty.
- FIR IDs: use format KA-[DIST]-[STN_SERIAL]-[YEAR]-[FIR_SERIAL] e.g. KA-MYS-012-2024-0347.
- Station names: use real KSP stations — Jayanagar, Indiranagar, Sadashivanagar, Vijayanagar, Yelahanka, Mysuru North, Dharwad, Hubli Town, Mangaluru East, Belagavi Camp.

- BNS sections: Theft BNS 303/IPC 379, Robbery BNS 309/IPC 392, House-breaking BNS 305/IPC 457, Assault BNS 115/IPC 323, Murder BNS 101/IPC 302, Cheating BNS 318/IPC 420. All FIRs before July 1 2024 must use IPC.
- Aliases: Give at least 20% of accused persons 2–3 name variants across different scripts and spellings. This is what makes entity resolution (C8) demonstrable.

Mandatory Demo Seeds

i These specific patterns must be seeded deliberately — they are the backbone of the live demo stories.

Story 1 – Entity resolution	One person with 4 aliases (English + Kannada + misspelling) appearing in 3 different stations across 2 districts.
Story 2 – MO match	One MO code appearing in 6 FIRs across 4 stations, with varied accused but identical method. At least 2 are solved.
Story 4 – Network sketch	One gang of 8 members linked across 12 FIRs in 3 districts; at least 3 shared phone numbers across members.
Story 5 – Deadline guard	3 FIRs with chargesheet deadlines within 7 days of the demo date, assigned to the demo IO officer.
Story 16 – Vehicle check	5 vehicles flagged as stolen with clearly recognisable Karnataka registration numbers.
Story 18 – Repeat victim	One address / household with 4 complaint records over 3 years — escalating in severity from NCR to FIR.
Story 19 – Chargesheet brief	One FIR with 8+ case diary entries, 3 accused, 2 witnesses, and both IPC history and a BNS chargesheet filed.
Story 21 – Bandobast	Dasara festival in Mysuru with 3 years of historical incidents seeded in the events_calendar and linked FIRs.

7. Phase 2 Scope (Not in v1)

The following data capabilities are designed for but not built in the first prototype:

Spatial queries (PostGIS)	lat/lng columns are stored; PostGIS extension and spatial query operators deferred.
Map rendering	Leaflet/Mapbox UI deferred; location shown as text/name groupings in v1.
Hotspot detection algorithms	KDE, DBSCAN, grid heatmaps deferred — require validated coordinates.
Ward-level external data	district_socioeconomic is district-level only; ward shapefile joins are Phase 2.
Financial transaction tables	Module 7 (Financial Crime) deferred entirely — no transaction tables in v1 schema.
Predictive model outputs	Module 8 (Forecasting) deferred — C11 component not built in v1.

— End of document —